



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE(Advanced)
REVIEW TEST
15-03-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE : TEAS]

Time : 3 Hours

PAPER-1

Maximum Marks : 180

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **28** pages and that all the **19** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140

- **Boltzmann constant** $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$
- **Coulomb's law constant** $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$
- **Universal gravitational constant** $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$
- **Speed of light in vacuum** $c = 3 \times 10^8 \text{ ms}^{-1}$
- **Stefan-Boltzmann constant** $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$
- **Wien's displacement law constant** $b = 2.89 \times 10^{-3} \text{ m-K}$
- **Permeability of vacuum** $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$
- **Permittivity of vacuum** $\epsilon_0 = \frac{1}{\mu_0 c^2}$
- **Planck constant** $h = 6.63 \times 10^{-34} \text{ J-s}$

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I (i) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the correct option is chosen.

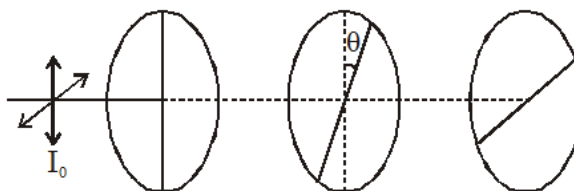
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

1. A capacitor is connected to an external circuit such that current in the circuit is to the left and decreasing. Right plate is positively charged. The displacement current in between the plates is :



- (A) to right and increasing. (B) to right and decreasing.
(C) to left and increasing. (D) to left and decreasing.
2. Two polarizing filters are crossed at 90° . A third filter is placed between them initially aligned with first. It is then rotated by angle θ . Find plot of intensity coming from third and final filter as a function of θ . The incident light is unpolarized with intensity I_0 :-



Space for Rough Work

3. If the wavelength of an electromagnetic radiation is of the order of size of a football field, which radiation is it ?
(A) Microwave (B) Gamma ray (C) Radio wave (D) x-ray
4. A horizontal force F acts at the centre of mass of a uniform disc with mass M and radius R . What is maximum value of F for which the disc can roll without slipping ; (μ_s is coefficient of friction) :-
(A) $\frac{2}{3}\mu_s Mg$ (B) $\frac{3}{2}\mu_s Mg$ (C) $2\mu_s Mg$ (D) $3\mu_s Mg$
-

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

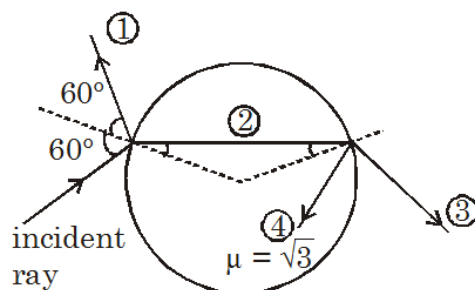
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

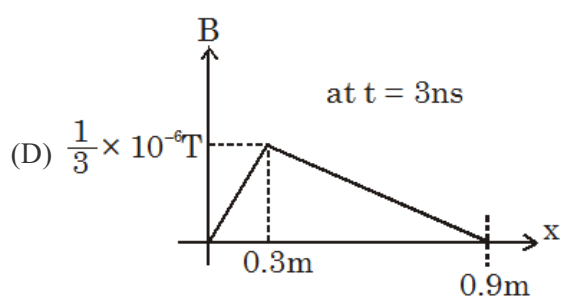
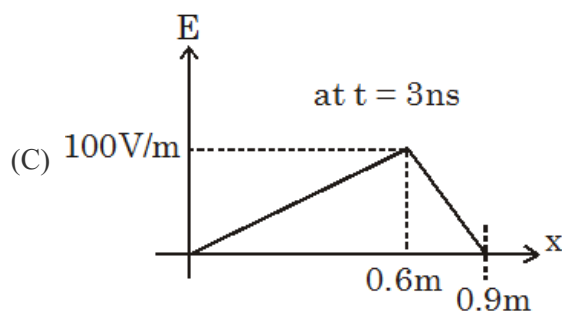
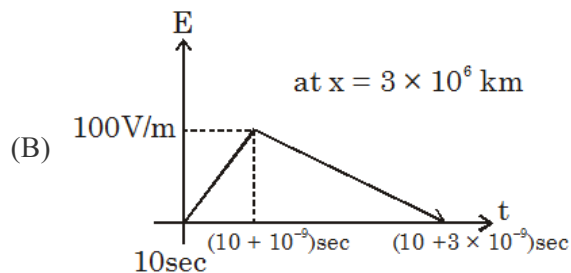
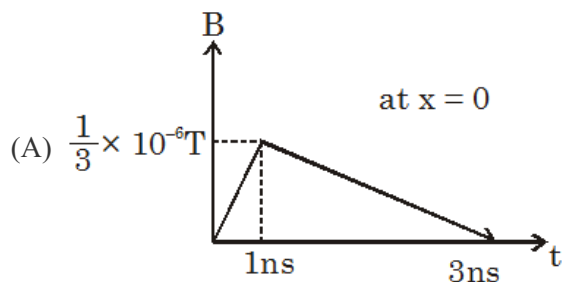
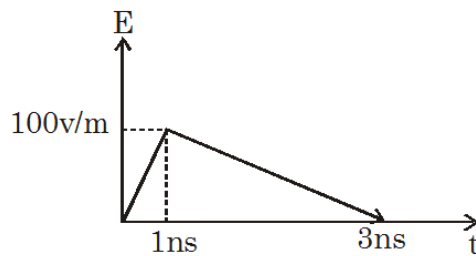
5. Consider an unpolarized light ray incident on a transparent sphere of refractive index $\sqrt{3}$ as shown. Which of the following statement (s) is/are correct ?



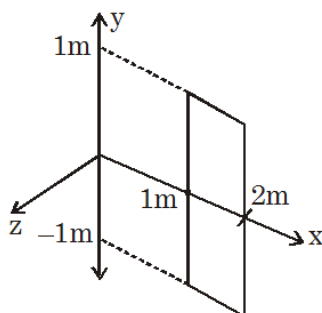
- (A) Ray 1 is completely polarized. (B) Ray 2 is partially polarized.
(C) Ray 4 is completely polarized. (D) Ray 3 is partially polarized.

Space for Rough Work

6. Consider a pulse of plane electromagnetic wave whose electric field is in y direction and is propagating in +x direction. The electric field at $x = 0$ varies as shown in the graph. Choose the correct graph (s) for electric field and magnetic field as x and t.



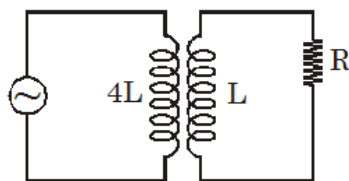
7. A plane electromagnetic wave $\vec{E} = 3000 \frac{\text{N}}{\text{C}} \hat{j} \sin(\pi x - 3\pi \times 10^8 t)$ is travelling in vacuum. A conducting loop of dimensions as shown is kept in xy plane as shown. The induced emf in the loop :-



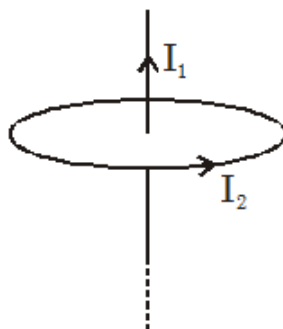
- (A) At $t = 0$ is 0
 (B) At $t = 0.01 \mu\text{s}$ is 12000 V
 (C) At $t = 0.003 \mu\text{s}$ is 12000 V
 (D) At $t = \frac{1}{1800} \mu\text{s}$ is 6000 V

Space for Rough Work

8. A current source carrying current $i = i_0 \sin \omega t$ is connected to primary of an ideal transformer. The primary has a self inductance $4L$ and secondary has self inductance L . There is no flux leakage. The resistance of primary is zero but resistance of secondary circuit is R as shown. (Take: $\omega L = R$) :-



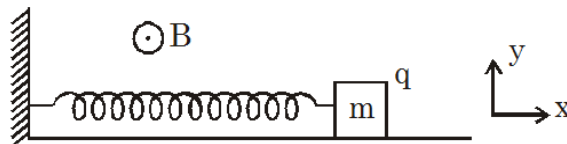
- (A) The current through secondary is $\sqrt{2}i_0 \cos\left(\omega t - \frac{\pi}{4}\right)$.
- (B) The voltage drop across resistance is $\sqrt{2}i_0 R \cos\left(\omega t - \frac{\pi}{4}\right)$.
- (C) The emf across primary is $\sqrt{2}i_0 R \sin\left(\omega t - \frac{\pi}{4}\right)$.
- (D) The emf across primary is $2i_0 R \cos\left(\omega t - \frac{\pi}{4}\right)$
9. An infinitely long, straight wire carrying current I_1 passes through the center of a circular loop of wire carrying current I_2 as shown in figure. The infinite wire is perpendicular to the plane of the loop. Which of the following statements are Incorrect regarding the magnetic force on the loop due to the infinite wire ?



- (A) Force on the loop due to infinite wire is upward, along the axis of the loop.
- (B) Force on the loop due to infinite wire is downward, along the axis of the loop.
- (C) Although net force on the loop due to infinite wire is zero, but due to magnetic interaction between infinite wire and loop, a tensile stress is developed in the loop.
- (D) There is no magnetic force on the loop due to infinite wire.

Space for Rough Work

10. A spring of spring constant k is fixed at one end has a small block of mass m and charge q is attached at the other end. The block rests over a smooth horizontal surface. A uniform and constant magnetic field B exists normal to the plane of paper as shown in figure. An electric field $\vec{E} = E_0 \hat{i}$ is switched on at $t = 0$. The block moves on the horizontal surface without lifting off the surface. Then the normal reaction acting on the the block is :-



- (A) Maximum at the mean position. (B) Minimum at the mean position.
 (C) Equal to mg at extreme position. (D) More than mg at extreme position.

Space for Rough Work

SECTION-II (i) : (Maximum Marks: 12)

- This section contains **THREE (03)** question stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value corresponding to the answer in the designated place.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:

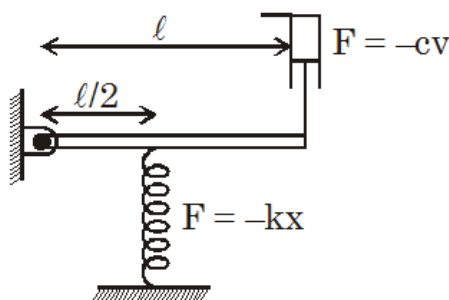
Full Marks : +2 If ONLY the correct numerical value is entered at the designated place;

Zero Marks : 0 In all other cases.

Paragraph for Questions 1 and 2

A uniform rod of mass m and length ℓ is free to rotate about an axis passing through its end. Neglect gravity. There is a spring of constant k at $\frac{\ell}{2}$ from axis and damper of $F = -cv$ at the free end. If rod is

displaced slightly, it executes damped oscillations. Take $\frac{k}{m} = \frac{25}{3} \frac{c^2}{m^2}$, $\frac{c}{m} = 10/\text{sec}$.



1. What is the angular frequency of these oscillations.
2. If the distance of spring from the hinge is $n\ell$, the motion is critically damped. Find n .

Space for Rough Work

Paragraph for Questions 3 and 4

A question arises that what is the ultimate source of radiant energy in case of electromagnetic waves. According to Maxwell's equation, this em wave is due to accelerated charged particles. For a non relativistic particle ($v \ll c$) it can be shown that power radiated by the charge particle is given by

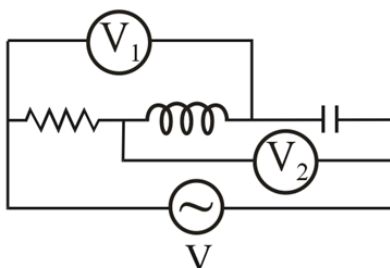
$$P = \frac{q^2 a^2}{6\pi\epsilon_0 c^3} \text{ where } q \text{ is charge, } a \text{ is acceleration, } c \text{ is speed of light.}$$

Once such radiation is emitted, it travels through space as a normal em wave. If charge is undergoing SHM, the electromagnetic wave is also harmonic of same frequency. In general, the radiation emitted by an accelerating charge is not isotropic. Let us consider a charge q oscillating with an angular frequency ω . It's damping constant b due to radiation losses is dependent on amplitude A as A^x and on angular frequency ω as ω^y .

3. The value of x is
4. The value of y is

Paragraph for Questions 5 and 6

In the RLC series circuit shown, the readings of voltmeters are $V_1 = 150 \text{ V}$, $V_2 = 50 \text{ V}$ and the source has emf 130 V .



5. The power factor of the circuit is $\frac{6n}{13}$ find n .
6. Value of $\frac{4V_L}{V_C}$ is:

Space for Rough Work

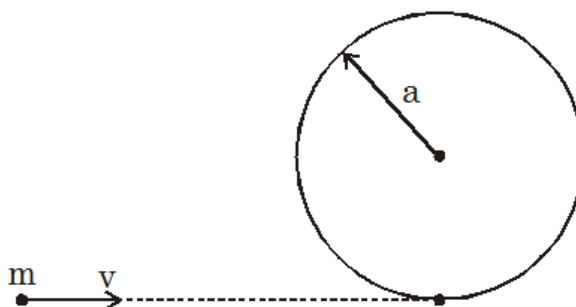
SECTION-II (ii) : (Maximum Marks: 12)

- This section contains **THREE (03)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +4 If only the correct answer is given.

Zero Marks : 0 In all other cases

7. A simple pendulum perform damped oscillations with logarithmic decrement = $\ln 8$. In how many complete oscillations will its maximum acceleration decrease by a factor of 64 ?
8. An electric dipole has a dipole moment $\vec{p} = \vec{p}_0 \sin \omega t$. The average power radiated by the dipole is $\frac{p_0^2 \omega^4}{n\pi\epsilon_0 C^3}$. Find value of n.
9. A uniform disc rests on a smooth horizontal table. It has mass m, radius a & centre C. A particle of mass m moving with speed v strikes at a point on the periphery and sticks to it. What is the loss in energy of the system. If loss in energy is $\frac{mv^2}{n}$, find value of n.



Space for Rough Work

PART-2 : CHEMISTRY
SECTION-I (i) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

1. 10 ml of 0.1 M of an acid was added to 50 ml of 0.1 M $\text{Na}_2\text{S}_2\text{O}_3$. The following readings were obtained :

$\ln K$	T(in K)
4	250K
5	333.33 K
6	500 K
7	1000 K

K is the rate constant in s^{-1} .

Find activation energy in kcal/mole. ($R = 2 \text{ cal/mol/K}$)

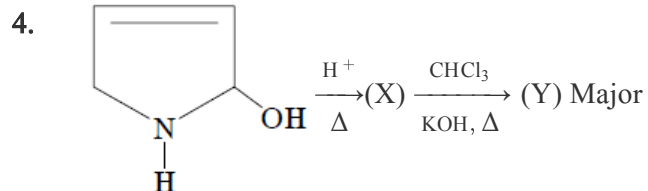
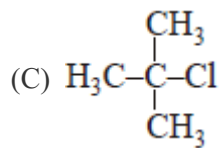
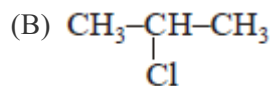
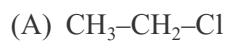
- (A) 1 (B) 1000 (C) 2 (D) 2000

2. In which of the following reaction bond cleavage of B_2H_6 is different from the other reactions.

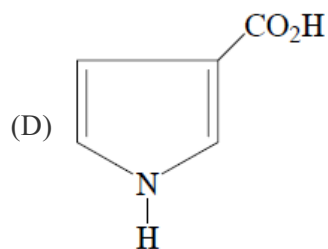
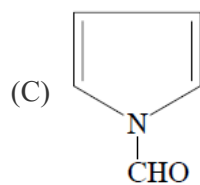
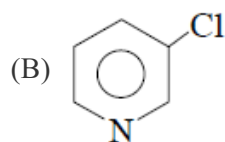
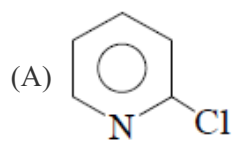
- (A) $\text{B}_2\text{H}_6 + 2\text{NMe}_3 \longrightarrow$ (B) $\text{B}_2\text{H}_6 + 2\text{THF} \longrightarrow$
(C) $\text{B}_2\text{H}_6 + 2\text{CO} \longrightarrow$ (D) $\text{B}_2\text{H}_6 + 2\text{NH}_3 \longrightarrow$

Space for Rough Work

3. Which alkyl halide is most reactive towards S_N1 as well as E_2 reaction ?



Product (Y) is



Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
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Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

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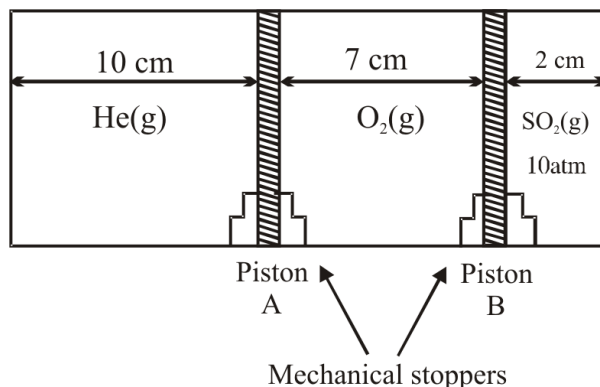
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- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

5. Equal mass of three gases He, O₂ & SO₂ are present in a container initially as shown -



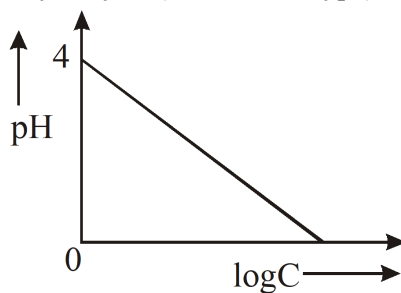
If mechanical stoppers are removed and movable pistons are allowed to attain equilibrium -

Choose the **incorrect** statement -

- (A) Position of piston A from the left end of the container is 15 cm
- (B) Position of piston B from the left end of the container is 7cm
- (C) Final pressure of O₂ gas is 10 atm
- (D) Final pressure of He gas is 20 atm

Space for Rough Work

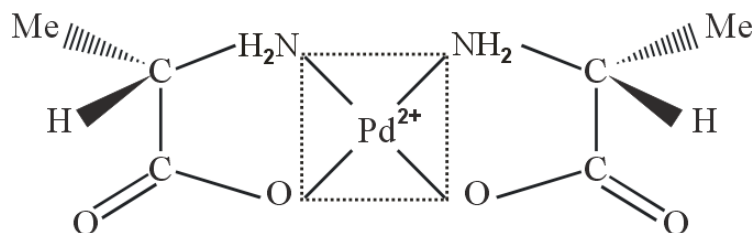
6. Based on the following graph for salt hydrolysis (monovalent type) mark the correct statement -



- (A) Salt may be NH_4Cl type
 (B) K_b of weak base is 10^{-6}
 (C) K_b of weak base is 10^{-3}
 (D) If above salt solution is titrated against strong base then phenolphthalein may be used as indicator
7. Which of the following statement(s) is/are **CORRECT** -
- (A) $[\text{B}_3\text{O}_6]^{-3}$ ion is non-planar but $\text{B}_3\text{N}_3\text{H}_6$ is planar
 (B) $(\text{H}_3\text{Si})_3\text{N}$ is planar but $(\text{SiH}_3)_3\text{P}$ is pyramidal
 (C) $\text{H}_3\text{C} - \text{NCS}$ is bent but $\text{SiH}_3 - \text{NCS}$ is linear
 (D) $(\text{CH}_3)_3\text{N}$ is pyramidal but $(\text{GeH}_3)_3\text{N}$ is planar

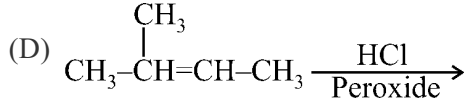
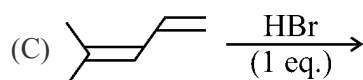
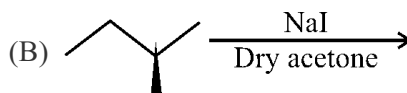
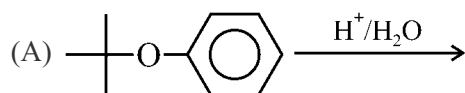
Space for Rough Work

8. One of the isomer of $[\text{Pd}(\text{NH}_2\text{--CH}(\text{CH}_3)\text{--CO}_2^-)_2]$ is given as



The correct statement(s) regarding the above compound is/are.

- (A) Total stereoisomers will be 6
 (B) The ligand $(\text{NH}_2\text{--CH}(\text{CH}_3)\text{--COO}^-)$ is Ambidentate ligand
 (C) The molecule is planar since Pd^{2+} shows square planar geometry around it
 (D) Exactly two Enantiomeric pair will be formed
9. In which of the following reaction(s), carbocation is involved as intermediate :



10. Which of the following species are electrophile :

- (A) Carbocation (B) Free radical (C) Carbene (D) Carbanion

Space for Rough Work

SECTION-II (i) : (Maximum Marks: 12)

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Paragraph for Questions 1 and 2

When 100 mL of 1.0 M HCl was mixed with 100 mL of 1.0 M NaOH in an insulated beaker at constant pressure, a temperature increase of 5.7 °C was measured for the beaker and its contents.

(Expt-1). Because the enthalpy of neutralisation of a strong acid with a strong base is a constant (-57.0 kJmol^{-1}), this experiment could be used to measure the calorimeter constant. In a second experiment (Expt-2), 100 mL of 2.0 M acetic acid ($K_a = 2.0 \times 10^{-5}$) was mixed with 100 mL of 1.5 M NaOH (under identical conditions to (Expt-1)) where a temperature rise of 8.25 °C was measured. (Consider heat capacity of all solutions as $4.2 \text{ Jg}^{-1}\text{K}^{-1}$ and density of all solutions as 1.0 g mL^{-1} , $\log 2 = 0.3$, $\log 3 = 0.48$)

1. Enthalpy of dissociation (in kJ mol^{-1}) of acetic acid obtained from the Expt-2 is
2. The pH of the solution after Expt-2

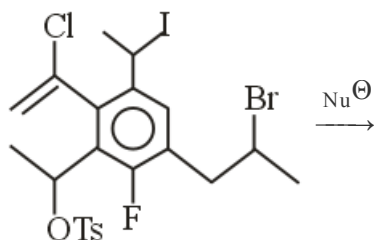
Space for Rough Work

Paragraph for Questions 3 and 4

Where two or more than two complexes having similar molecular weight (or) molecular formula but their structural arrangements are different these forms are known as isomer & this phenomenon is known as isomerism; isomerism are of two types structural & stereoisomerism ; stereoisomerism are of two types geometrical isomerism & optical isomerism; compound which has plane of symmetry will not exhibit optical isomerism.

3. Total possible isomers of $K_4[Fe(CN)_6]$.
4. Total geometrical isomers theoretically possible in an octahedral complex of formula $[Co(C_2O_4)(dmg)(en)]$ (In this complex dmg^{-1} is bonded by both 'N' atoms with central metal)

Paragraph for Questions 5 and 6



5. How many groups substituted by Nu[⊖], via S_N mechanism?
6. Number of groups can not show nucleophilic substitution reaction

Space for Rough Work

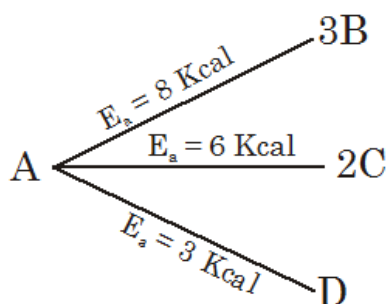
SECTION-II (ii) : (Maximum Marks: 12)

- This section contains **THREE (03)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +4 If only the correct answer is given.

Zero Marks : 0 In all other cases

7. A substance undergoes first order parallel reaction as shown. Calculate net activation energy (in kcal) if it is known that B, C and D are formed in a molar ratio of 2 : 3 : 4 respectively.



Express answer in nearest integer.

8. Which of the following compounds/species are incorrectly matched against the indicated names.
- $[\text{PdI}_2(\text{ONO})_2(\text{H}_2\text{O})_2] \rightarrow$ Diaquadiiodidodinitrito-Npalladium(IV)
 - $\text{Fe}(\pi\text{-C}_5\text{H}_5)_2 \rightarrow$ Bis(cyclopentadienyl) iron(II)
 - $\text{H}_2[\text{PtCl}_6] \rightarrow$ Hexachloridoplatinate(IV) hydride
 - $[\text{Al}(\text{OH})(\text{H}_2\text{O})_5]\text{SO}_4 \rightarrow$ Pentaquahydroxidoaluminium(III) sulphate
 - $\text{K}_2[\text{OsCl}_5\text{N}] \rightarrow$ Potassium nitridopentachloridoosmate(VI)
 - $\text{Cr}(\text{C}_6\text{H}_6)_2 \rightarrow$ Bis(η^6 -benzene)chromium(0)
9. An eq. solution of 0.2 N M $(\text{NO}_3)_2$ is saturated with 0.34% w/v H_2S . At what maximum pH precipitation of MS can be prevented.
- Given $(K_{sp})_{\text{MS}} = 10^{-19}$, $(K_{a1})_{\text{H}_2\text{S}} = 10^{-7}$, $(K_{a2})_{\text{HS}^-} = 10^{-14}$

Space for Rough Work

PART-3 : MATHEMATICS
SECTION-I (i) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

-
1. The ordinate of a point describing the circle $x^2 + y^2 = 25$ decreases at the rate of 1 cm/sec, then the rate of change (in cm/sec) of abscissa of the point when ordinate equal to 3cm, is-
(Given $x > 0, y > 0$)
(A) 0.25 (B) 0.5 (C) 0.75 (D) 1
 2. $\lim_{x \rightarrow -\infty} \left(\sqrt{x^2 - \frac{4x}{3}} + x + 2 \right)$ is equal to-
(A) $\frac{4}{3}$ (B) $-\frac{4}{3}$ (C) $\frac{8}{3}$ (D) $-\frac{8}{3}$
 3. If a derivable function $y = f(x)$ satisfies the condition $f(x) + f(y) = f(x)f(y) + f(xy) \forall x, y \in \mathbb{R}$ where $f(1) = 0$ and $f'(1) = -2$, then the number of solution(s) of the equation $f(x) = 2^x$ is
(A) 4 (B) 3 (C) 2 (D) 1
 4. Let $f(x)$ be a twice differentiable function satisfying $xf(x^2) + (x^2 + 1)f(x) = x, \forall x \in \mathbb{R}$ then the value of $f''(1)$ is
(A) $\frac{1}{3}$ (B) $-\frac{1}{3}$ (C) $-\frac{1}{9}$ (D) 1
-

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

5. Let $f(x) = 2^x + 4^x + 8^x$ and $g(x)$ is inverse function of $f(x)$, then (where $[.]$ represents greatest integer function)-

(A) $g'(14) \ln 2 < 1$

(B) $[g'(14) \ln 2] = 0$

(C) $g'(14) \ln 2 > 1$

(D) $[g'(14) \ln 2] = 1$

6. Let $y = f(x)$ is a positive valued function which satisfies equation $\sqrt{y^2 + 2x} + \sqrt{y^2 - 2x} = 2x^2$, then $\frac{dy}{dx}$ is equal to-

(A) $\frac{2x^3 - x^{-3}}{y}$

(B) $\frac{2x^4 - x^{-2}}{y}$

(C) $\frac{2x^4 - x^{-2}}{\sqrt{x^6 + 1}}$

(D) $\frac{2x^3 - x^{-3}}{\sqrt{1 + x^6}}$

Space for Rough Work

7. Let $f : \left[-\frac{1}{3}, 3\right] \rightarrow \mathbb{R}$ and $g : \left[-\frac{1}{3}, 3\right] \rightarrow \mathbb{R}$ be function defined by $f(x) = [x^2 - 4]$ and $g(x) = |x-2| f(x) + |3x-5| f(x)$, where $[y]$ denotes the greatest integer less than or equal to y for $y \in \mathbb{R}$. Then
- (A) f is discontinuous exactly at eight points in $\left[-\frac{1}{3}, 3\right]$
- (B) f is discontinuous exactly at nine points in $\left[-\frac{1}{3}, 3\right]$
- (C) g is NOT differentiable exactly at ten points in $\left(-\frac{1}{3}, 3\right)$
- (D) g is NOT differentiable exactly at nine points in $\left(-\frac{1}{3}, 3\right)$
8. If the solution of the equation, $\sin(\tan^{-1}x) = \sqrt{4 - (\sin(\cos^{-1}x) + \cos(\sin^{-1}x))^2}$ is 'a', then
- (A) $\sin^{-1}a + \cos^{-1}a = \frac{\pi}{2}$
- (B) $2\sin^{-1}a + \cos^{-1}a = \frac{\pi}{2}$
- (C) $\sin^{-1}a + 3\cos^{-1}a = \frac{3\pi}{2}$
- (D) $\tan^{-1}a + \cos^{-1}a = \frac{\pi}{2}$

Space for Rough Work

9. If a function $f : \mathbb{R} \rightarrow \mathbb{R}$ satisfies $\left| \sum_{k=1}^n 3^k (f(x+ky) - f(x-ky)) \right| \leq 1$ for every $n \in \mathbb{N}$ and for all $x, y \in \mathbb{R}$, then which of the following is/are true

(A) $|3^n (f(x+ny) - f(x-ny))| \leq 2$

(B) $|f(u) - f(v)| \leq \frac{2}{3^n}$

(C) $f(x)$ is a constant function

(D) $f'(x) = 1 \forall x \in \mathbb{R}$

10. Which of the following satisfy the equation $\sin^{-1} \sqrt{\frac{1+x}{2}} - \sqrt{2-x} = \cot^{-1} (\tan \sqrt{2-x}) - \sin^{-1} \sqrt{\frac{1-x}{2}}$

(A) $\left(2 - \frac{\pi^2}{4}, 1\right]$

(B) $\left(2 - \frac{\pi^2}{4}, 0\right)$

(C) $[-1, 1]$

(D) $\left[1 - \frac{\pi^2}{4}, 1\right]$

Space for Rough Work

SECTION-II (i) : (Maximum Marks: 12)

- This section contains **THREE (03)** question stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value corresponding to the answer in the designated place.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +2 If ONLY the correct numerical value is entered at the designated place;

Zero Marks : 0 In all other cases.

Paragraph for Questions 1 and 2

Let $g : \mathbb{R} \rightarrow \left\{ \frac{1}{2} \right\}$ be a function given by $g(x) = x^{10}(f'(t) - 2) + x^9(f''(t)) + 4xf(0) + \frac{1}{2}$ where $t \in \mathbb{R}$.

1. Number of solutions of $|f(x)| = \cos x$, is -
2. Number of points of non differentiability of function $h(x) = |f(x)(e^{f(x)} - 1)(f(x) - 1)(f(x) - 2)|$, is

Paragraph for Questions 3 and 4

If $y = \cot^{-1} \left(\frac{\sqrt{1 + \sin x} + \sqrt{1 - \sin x}}{\sqrt{1 + \sin x} - \sqrt{1 - \sin x}} \right)$, then

3. For $x \in \left(0, \frac{\pi}{2} \right)$, $\frac{dy}{dx}$ is equal to
4. For $x \in \left(\frac{\pi}{2}, \pi \right)$, $\frac{dy}{dx}$ is equal to

Space for Rough Work

Paragraph for Questions 5 and 6

Given two curves : $C_1 : y = x^3 + 3x^2$ and $C_2 : y = f(x)$ (where $f(x)$ is a differentiable function)

5. If equation of normal to the curve $y = f(x)$ at $x = 1$ is $3x + 2y + 1 = 0$, then $\lim_{x \rightarrow 1} \frac{(e^{\sin \pi x} - 1) \cdot \sin \frac{\pi}{x}}{\pi^2 \tan^{-1}(x-1) \cdot (f^2(x^3) - 4)}$ is equal to
6. The value of 9λ such that we can draw three distinct tangents to the curve C_1 from point $(\lambda, 0)$ such that two tangents are perpendicular to each other is -

Space for Rough Work

SECTION-II (ii) : (Maximum Marks: 12)

- This section contains **THREE (03)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +4 If only the correct answer is given.

Zero Marks : 0 In all other cases

7. $\lim_{x \rightarrow 0^+} (x \cos x)^x + (x \sin x)^{1/x}$ is
8. Let $4 \left(\frac{ay}{\cos \alpha} - \frac{bx}{\sin \alpha} \right) = 3(a + b)$, $ab \neq 0$ is normal to the curve $x = 3\cos\theta - \cos^3\theta$, $y = 3\sin\theta - \sin^3\theta$ at $\theta = \alpha$, then number of possible values of α , $\alpha \in (0, \pi) - \left\{ \frac{\pi}{2} \right\}$ is
9. Number of point(s) in $[0, 9]$, where $f(x) = \{x\} \left| \sin \frac{\pi x}{2} \right| - \left\{ \frac{x}{2} \right\}$ is discontinuous, is
(where $\{.\}$ denotes fractional part function)

Space for Rough Work

Space for Rough Work

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions
and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form
number of the candidate, and that question
paper and ORS codes are the same.

Signature of the Invigilator

Space for Rough Work